

Ex# 4.7

$$Q \quad 1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{512}$$

Here,

$$a=1, \quad r=\frac{1}{2} = \frac{1}{2} < 1$$

$$\therefore S_n = \frac{a(1-r^n)}{1-r} \rightarrow (i)$$

First we find 'n'

$$a_n = ar^{n-1}$$

$$\frac{1}{512} = (1)\left(\frac{1}{2}\right)^{n-1}$$

$$\left(\cancel{\frac{1}{2}}\right)^9 = \left(\cancel{\frac{1}{2}}\right)^{n-1}$$

$$1 + 9 = n$$

$$\boxed{n = 10}$$

$$\Rightarrow S_{10} = \frac{1}{1 - \frac{1}{2}} \left\{ 1 - \left(\frac{1}{2}\right)^{10} \right\}$$

$$\Rightarrow S_{10} = \frac{1023}{512}$$

Ex# 4.7

Date _____

Q1

(iii)

$$\frac{1}{n} + \frac{1}{n^2} + \frac{1}{n^3} + \dots + \frac{1}{n^8}$$

$$S_n = ?$$

Here

$$a = \frac{1}{n}, \quad r = \frac{1}{n} = \frac{1}{\frac{1}{n^2}}$$

$$a_n = \frac{1}{n^8}, \quad n = 8$$

$$\therefore S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_8 = \frac{\frac{1}{n} \left\{ \left(\frac{1}{n} \right)^8 - 1 \right\}}{\left(\frac{1}{n} \right) - 1}$$

$S_8 \Rightarrow$

$$\frac{\frac{1}{n} \left\{ \left(\frac{1}{n} \right)^8 - 1 \right\}}{\frac{1 - n}{n}}$$

$$S_8 = \frac{\left(\frac{1 - n^8}{n^8} \right)}{1 - n}$$

$$S_8 = \left(\frac{1 - n^8}{n^8 (1 - n)} \right)$$

iv) $0.9 + 0.99 + 0.999 + \dots$ to n terms

$$\Rightarrow \frac{9}{10} + \frac{99}{100} + \frac{999}{1000} + \dots \text{ to } n \text{ terms}$$

$$\Rightarrow \left(1 - \frac{1}{10} \right) + \left(1 - \frac{1}{100} \right) + \left(1 - \frac{1}{1000} \right) + \dots \text{ to } n \text{ terms}$$

$$\Rightarrow (1 + 1 + 1 + \dots \text{ to } n \text{ terms}) - \left(\frac{1}{10} + \frac{1}{100} + \frac{1}{1000} + \dots \text{ to } n \text{ terms} \right)$$

$$\Rightarrow n - \left(\text{G.P with } a = \frac{1}{10}, r = \frac{1}{10}, n = n \right)$$

$$\therefore S_n = \frac{a(1 - r^n)}{1 - r}, r < 1$$

$$\Rightarrow 8n - \left\{ \frac{\left(\frac{1}{10}\right) \cdot \left\{1 - \left(\frac{1}{10}\right)^n\right\}}{1 - \frac{1}{10}} \right\}$$

$$\Rightarrow n - \left\{ \frac{\frac{1}{10} \left\{1 - \left(\frac{1}{10}\right)^n\right\}}{\frac{10-1}{10}} \right\}$$

$$\Rightarrow n - \left\{ \frac{\left\{1 - \left(\frac{1}{10}\right)^n\right\}}{9} \right\}$$

$$\Rightarrow n - \left[\frac{1 - \frac{1}{10^n}}{9} \right]$$

$$\Rightarrow n - \left[\frac{\frac{10^n - 1}{10^n}}{9} \right]$$

$$\Rightarrow n - \left[\frac{10^n - 1}{9(10^n)} \right]$$

$$\Rightarrow n - \frac{1}{9} \left[\frac{10^n - 1}{10^n} \right]$$

$$\Rightarrow n - \frac{1}{9} \left[1 - \frac{1}{10^n} \right] \quad \text{Ans}$$

v) $7 + 77 + 777 + \dots$ to n terms.

$$7 (1 + 11 + 111 + \dots \text{ to } n \text{ terms})$$

multiply and divide by 9.

$$\Rightarrow \frac{7}{9} (9 + 99 + 999 + \dots \text{ to } n \text{ terms})$$

$$\Rightarrow \frac{7}{9} \left\{ (10-1) + (100-1) + (1000-1) + \dots \right\} \text{ to } n \text{ terms}$$

$$\Rightarrow \frac{7}{9} \left\{ (10 + 100 + 1000 + \dots \text{ to } n \text{ terms}) - (1 + 1 + 1 \text{ to } n \text{ terms}) \right\}$$

$$\Rightarrow \frac{7}{9} \left\{ (G.P. \text{ with } a=10, r=10, n=n) - n \right\}$$

$$\therefore S_n = \frac{a(r^n - 1)}{r - 1}$$

$$\Rightarrow \frac{7}{9} \left\{ \frac{10(10^n - 1)}{10 - 1} - n \right\}$$

$$\Rightarrow \frac{7}{9} \left\{ \frac{10(10^n - 1)}{9} - n \right\} \text{ Ans}$$

Q2 Find the sum to n terms of the following series :

$$i) 1 + (a+b) + (a^2+ab+b^2) + (a^3+a^2b+ab^2+b^3) + \{ \dots ;$$

where,

$$a > 1, b > 1$$

$$a > b$$

Multiply and divide by $(a-b)$

$$\frac{1}{(a-b)} \left\{ (a-b) + (a-b)(a+b) + (a-b)(a^2+ab+b^2) + \dots \right\}$$

$$\Rightarrow \frac{1}{a-b} \left\{ (a-b) + a^2 - b^2 + a^3 - b^3 + \dots \right\}$$

$$\Rightarrow \frac{1}{a-b} \left\{ (a + a^2 + a^3 + \dots) - (b + b^2 + b^3 + \dots) \right\}$$

$$\frac{1}{a-b} \left\{ \left(\text{GP with } a=a, r=a, n=n \right) \right.$$

$$\left. - \left(\text{GP with } a=b, r=b, n=n \right) \right\}$$

$$\frac{1}{a-b} \left\{ \frac{a(a^n-1)}{a-1} - \frac{b(b^n-1)}{b-1} \right\}$$

$$\Rightarrow \frac{1}{a-b} \left\{ \frac{a^{n+1}-a}{a-1} - \frac{b^{n+1}-b}{b-1} \right\} \text{ Ans}$$

Q3 If $a_n = \left(\frac{1}{4}\right)^n$ then find its sum upto n terms

put $n=1$
 $a_1 = \left(\frac{1}{4}\right)^1$

$$\boxed{a_1 = \frac{1}{4}}$$

put $n=3$
 $a_3 = \left(\frac{1}{4}\right)^3$

$$\boxed{a_3 = \frac{1}{64}}$$

put $n=2$
 $a_2 = \left(\frac{1}{4}\right)^2$

$$\boxed{a_2 = \frac{1}{16}}$$

$$r = \frac{a_2}{a_1}$$

$$r = \frac{1}{\frac{1}{4}}$$

$$\boxed{r = \frac{1}{4} < 1}$$

$$\therefore S_n = \frac{a(1-r^n)}{1-r}, r < 1$$

$$S_n = \left(\frac{1}{4}\right) \left[1 - \left(\frac{1}{4}\right)^n \right]$$

$$S_n = \frac{\left(\frac{1}{4}\right) \left\{ 1 - \left(\frac{1}{4}\right)^n \right\}}{\frac{4-1}{4}}$$

$$\boxed{S_n = \frac{1 - \left(\frac{1}{4}\right)^n}{3}}$$

Q₄ Find the sum of the following infinite geometric series.

$$i) \frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \dots$$

Here,

$$a = \frac{1}{5}, \quad r = \frac{1}{25} = \frac{1}{5}$$

$$\frac{1}{5}$$

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$$S_{\infty} = \frac{a}{1-r}$$

$$S_{\infty} = \frac{\frac{1}{5}}{1 - \left(\frac{1}{5}\right)} = \frac{\frac{1}{5}}{\frac{5-1}{5}}$$

$$\Rightarrow \boxed{S_{\infty} = \frac{1}{4}}$$

$$\text{ii)} \quad 2 + 2\sqrt{2} + 1 + \dots$$

$$a = 2,$$

$$r = \frac{1}{\sqrt{2}}$$

$$S_{\infty} = \frac{a}{1-r}$$

$$S_{\infty} = \frac{2}{1 - \frac{1}{\sqrt{2}}}$$

$$S_{\infty} = \frac{2}{\sqrt{2} - 1}$$

$$\boxed{S_{\infty} = \frac{2\sqrt{2}}{\sqrt{2}-1}}$$

$$\text{iii)} \quad 0.1 + 0.05 + 0.025 + \dots$$

$$a = 0.1 = \frac{1}{10}, \quad r = \frac{0.05}{0.1}$$

$$r = 0.5 = \frac{1}{2}$$

$$S_{\infty} = \frac{a}{1-r} = \frac{\left(\frac{1}{10}\right)}{1 - \frac{1}{2}}$$

$$S_{\infty} = \frac{\frac{1}{10}}{\frac{2-1}{2}}$$

$$S_{\infty} = \frac{1}{5}$$

Question (1)

vi) $0.9 + 0.22 + 0.222 + \dots$ to n terms

Sol:-

$$\Rightarrow 2 \left(0.1 + 0.11 + 0.111 + \dots \text{ to } n \text{ terms} \right)$$

multiply and divide by 9

$$\Rightarrow \frac{2}{9} \left(0.9 + 0.99 + 0.999 + \dots \text{ to } n \text{ terms} \right)$$

$$\Rightarrow \frac{2}{9} \left\{ \left(1 - \frac{1}{10} \right) + \left(1 - \frac{1}{100} \right) + \left(1 - \frac{1}{1000} \right) + \dots \text{ to } n \text{ terms} \right\}$$

$$\Rightarrow \frac{2}{9} \left\{ (1 + 1 + 1 + \dots \text{ to } n \text{ terms}) - \left(\frac{1}{10} + \frac{1}{100} + \frac{1}{1000} + \dots \text{ to } n \text{ terms} \right) \right\}$$

$$\Rightarrow \frac{2}{9} \left\{ n - \left(\text{GP with } a = \frac{1}{10}, r = \frac{1}{10}, n = n \right) \right\}$$

$$\therefore S_n = \frac{a(1-r^n)}{1-r}$$

Date _____

$$\Rightarrow \frac{2}{9} \left\{ n - \frac{\left(\frac{1}{10}\right)\left(1 - \left(\frac{1}{10}\right)^n\right)}{1 - \left(\frac{1}{10}\right)} \right\}$$

$$\Rightarrow \frac{2}{9} \left\{ n - \frac{\left(\frac{1}{10}\right)\left(1 - \left(\frac{1}{10}\right)^n\right)}{\frac{10-1}{10}} \right\}$$

$$\Rightarrow \frac{2}{9} \left\{ n - \frac{1 - \left(\frac{1}{10}\right)^n}{9} \right\}$$

$\Rightarrow$

$$\Rightarrow \frac{2}{9} \left\{ n - \frac{1}{9} \left\{ 1 - (0.1)^n \right\} \right\} \text{ Ans}$$

vii) $3 + 33 + 333 + \dots$ to n terms.

Sol:-

$$3 (1 + 11 + 111 + \dots \text{ to } n \text{ terms})$$

multiply and divide by 9

$$\frac{3}{9} (9 + 99 + 999 + \dots \text{ to } n \text{ terms})$$

$$\frac{3}{9} \left\{ (10-1) + (100-1) + (1000-1) + \dots \text{ to } n \text{ terms} \right\}$$

$$\Rightarrow \frac{1}{3} \left\{ (10 + 100 + 1000 + \dots \text{to } n \text{ terms}) - (1 + 1 + \dots \text{to } n \text{ terms}) \right\}$$

$$\Rightarrow \frac{1}{3} \left\{ (\text{GP with } a=10, r=10, n=n) - n \right\}$$

$$\therefore S_n = \frac{a(r^n - 1)}{r - 1}$$

$$\Rightarrow \frac{1}{3} \left\{ \frac{(10)(10^n - 1)}{10 - 1} - n \right\}$$

$$\Rightarrow \frac{1}{3} \left\{ \frac{10^{n+1} - 10}{9} - n \right\}$$

$$\Rightarrow \frac{1}{3} \left\{ \frac{10^{n+1} - 10 - 9n}{9} \right\}$$

$$\Rightarrow \frac{1}{27} (10^{n+1} - 10 - 9n) \quad \text{Ans}$$

ii) $r + (1+k)r^2 + (1+k+k^2)r^3 + \dots$

where k and r are proper fractions.

multiply and divide by $(1-k)$

$$S_n = \frac{1}{1-k} \left\{ (1-k^2)r + (1-k^2)r^2 + (1-k^3)r^3 + \dots \right\}$$

$$S_n = \frac{1}{1-k} \left\{ (r - kr) + (r^2 - k^2r^2) + (r^3 - k^3r^3) + \dots \right\}$$

$$S_n = \frac{1}{1-k} \left\{ (r + r^2 + r^3 + \dots \text{ to } n \text{ terms}) - (kr + k^2r^2 + k^3r^3 + \dots \text{ to } n \text{ terms}) \right\}$$

$$S_n = \frac{1}{1-k} \left\{ (\text{G.P. with } a=r, r=r, n=n) - (\text{G.P. with } a=kr, r=kr, n=n) \right\}$$

$$S_n' = \frac{a(1-r^n)}{1-r}$$

$$\Rightarrow S_n = \frac{1}{1-k} \left[\frac{r(1-r^n)}{1-r} - \frac{rk(1-r^n k^n)}{1-r} \right]$$

$$S_n = \frac{r}{1-k} \left[\frac{(1-r^n)}{1-r} - \frac{k(1-r^n k^n)}{1-r} \right] \text{ Ans.}$$

Q5

iv) 1.148

$$1.148 = 1 + \frac{1}{10} + (0.048 + 0.00048 + 0.0000048)$$

$$= 1 + \frac{1}{10} + \left(G \cdot P \text{ with } a = 0.048, r = 0.01 = \frac{1}{10} \right)$$

$n = \infty$

$$1 + \frac{1}{10}$$

$$\therefore S_{\infty} = \frac{a}{1-r}$$

$$S_{\infty} = 1 + \frac{1}{10} + \left(\frac{\frac{48}{1000}}{1 - \frac{1}{100}} \right)$$

$$S_{\infty} = 1 + \frac{1}{10} + \frac{48/1000}{100 - 1/100}$$

$$S_{\infty} = 1 + \frac{1}{10} + \frac{48}{10 \times 99}$$

$S_{\infty} =$	379
	330

Q. $a = \frac{b}{2} + \frac{b^2}{4} + \frac{b^3}{8} + \dots$, prove $b = \frac{2a}{1+a}$

Here

$$a' = \frac{b}{2}, \quad r = \frac{b^2}{4} = \frac{b}{2}$$

$$\frac{b}{2}$$

$$n = \infty$$

$$S_{\infty} = \frac{a}{1-r}$$

$$S_{\infty} = \frac{b}{2}$$

$$1 - \frac{b}{2}$$

$$S_{\infty} = \frac{b}{2} = \frac{b}{2-b}$$

Here,

$$a = \frac{b}{2-b}$$

$$a(2-b) = b$$

$$b = 2a - ab$$

$$b + ab = 2a$$

$$b(1+a) = 2a$$

$$b = \frac{2a}{1+a}$$

Q7

Sol:-

$$G.P = a + ar + ar^2 + \dots$$

$$G.P = a^2 + a^2r^2 + a^2r^4 + \dots$$

$$\text{Sum of infinite series} = \frac{1}{2} (\text{sum of square of its term}) \rightarrow (i)$$

Second condition,

$$a + ar = \frac{9}{2}$$

$$a(1+r) = \frac{9}{2}$$

$$a(1+r) = \frac{9}{2}$$

$$a = \frac{9}{2(1+r)} \rightarrow (ii)$$

$$\therefore r = \frac{a_2}{a_1} = \frac{ar}{a} = r$$

$$r = r$$

$$\therefore S_{\infty} = \frac{a}{1-r}$$

put in eq (i)

$$\Rightarrow \frac{a}{1-r} = \frac{1}{2} \left[\frac{a^2}{1-r^2} \right]$$

$$\Rightarrow \frac{a}{\cancel{1-r}} = \frac{1}{2} \left[\frac{a^2}{(1+r)\cancel{(1-r)}} \right]$$

$$\Rightarrow a = \frac{a^2}{2(1+r)}$$

$$\Rightarrow \frac{a}{a^2} = \frac{1}{2(1+r)} \Rightarrow \frac{1}{a} = \frac{1}{2(1+r)}$$

$$\Rightarrow a = 2(1+r) \Rightarrow \frac{a}{2(1+r)}$$

$$\Rightarrow \frac{a}{4} = (1+r)(1+r)$$

$$\Rightarrow \frac{a}{4} = 1^2 + 2(1)(r) + r^2$$

$$\Rightarrow a = 4 + 8r + 4r^2 \Rightarrow 4r^2 + 8r + 4 - a = 0$$

$$\Rightarrow 4r^2 + 8r - 5 = 0$$

$$\Rightarrow 4r^2 + 10r - 2r - 5 = 0$$

$$2r(2r+5) - 1(2r+5) = 0$$

$$(2r+5)(2r-1) = 0$$

equality method:-

$$2r+5=0$$

$$2r = -5$$

$$r = \frac{-5}{2}$$

$$2r-1=0$$

$$2r = 1$$

$$r = \frac{1}{2}$$

Not acceptable

put r in eq (ii)

$$a = \frac{9}{2(1+\frac{1}{2})}$$

$$a = \frac{9}{2(\frac{3}{2})}$$

$$a = \frac{9^3}{3} \Rightarrow a = 3$$

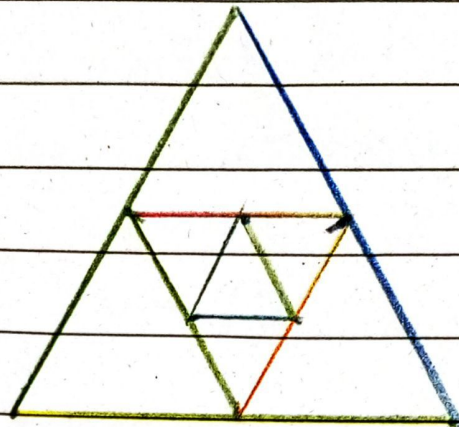
put 'a' and 'r' GP series:-

$$3 + (3)(1/2) + (3)(1/2)^2 + \dots$$

$$3 + \frac{3}{2} + \frac{3}{4} + \dots$$

Ans

Q.8



$$\text{Perimeter of original triangle} = \frac{5}{2}$$

$$\text{Perimeter of 1st nested triangle} = \frac{5}{2} = \frac{5}{4}$$

$$\text{Perimeter of 2nd nested triangle} = \frac{5}{4} = \frac{5}{8}$$

$$\frac{5}{2}, \frac{5}{4}, \frac{5}{8}, \dots$$

$$a = \frac{5}{2}, \quad r = \frac{5/4}{5/2} = \frac{1}{2}$$

$$\Rightarrow S_{\infty} = \frac{a}{1-r}$$

$$\Rightarrow S_{\infty} = \frac{5}{2}$$

$$\Rightarrow S_{\infty} = \frac{5}{2}$$

$$\Rightarrow \boxed{S_{\infty} = 5}$$

$$1 - \frac{1}{2}$$

$$\frac{1}{2}$$